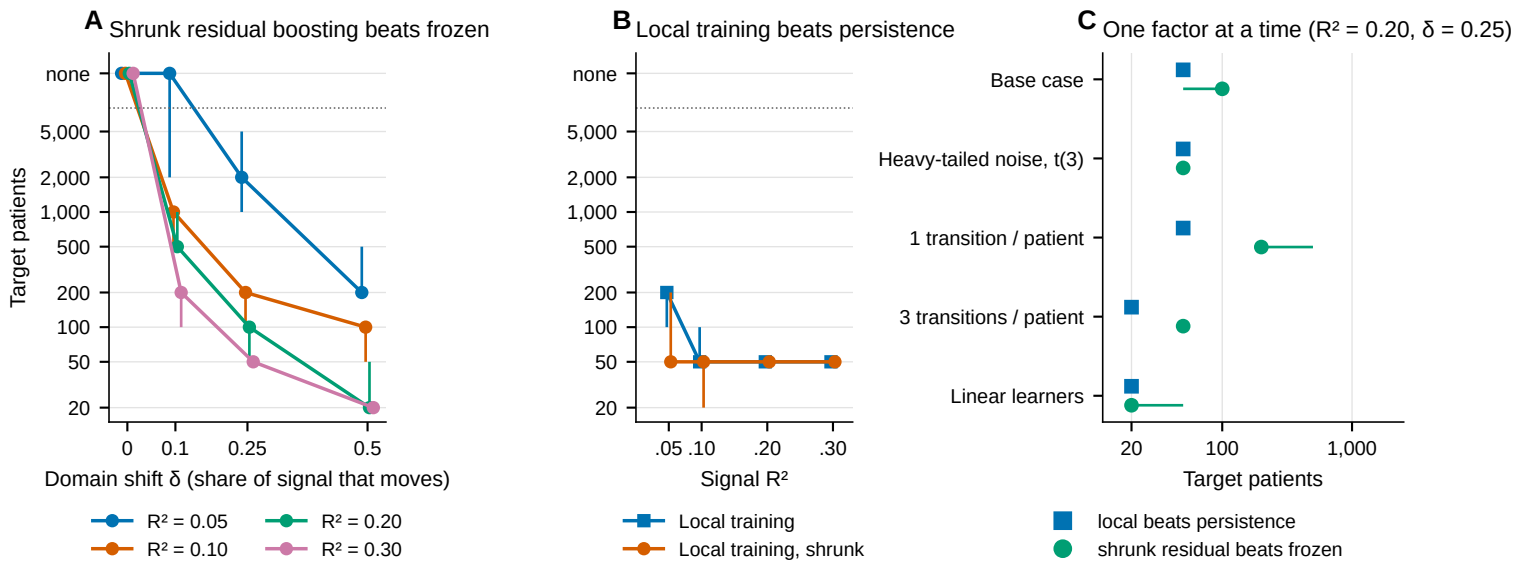
A Shrunk residual boosting beats frozen
Target patients
none
5,000
2,000
1,000
500
200
100
50
20
0
0.1
0.25
0.5
Domain shift δ (share of signal that moves)
R² = 0.05
R² = 0.10
R² = 0.20
R² = 0.30
B Local training beats persistence
none
5,000
2,000
1,000
500
200
100
50
20
.05
.10
.20
.30
Signal R²
Local training
Local training, shrunk
C One factor at a time (R² = 0.20, δ = 0.25)
Base case
Heavy-tailed noise, t(3)
1 transition / patient
3 transitions / patient
Linear learners
20
100
1,000
Target patients
local beats persistence
shrunk residual beats frozen